



Describe Systems Theory with respect to Information Systems

The business environment, the management system, and the information systems that serve every level of the organisation

TIME	SESSION	MODULE	QUALITY ASSURANCE
90 minutes · Self & Group	Friday, 21 Aug 2026 · 09h00 – 14h00	Module 1 · Professional Team Development	QCTO / MICT SETA

The manager and the management system

- A manager is someone skilled in knowing how to analyse and improve the ability of an organisation to survive and grow in a complex and changing world.
- Managers have a set of tools that enable them to grasp the complexity of the organisation's environment.
- A management system describes the organisation and the set of significant interacting institutions and forces in its complex, rapidly changing environment that affect its ability to serve its customers.
- The firm must continuously monitor and adapt to the environment if it is to survive and prosper.
- Disturbances in the environment may spell profound threats or new opportunities — the successful firm identifies, appraises and responds to both.

TWO LEVELS

Internal and external environment



Internal environment

The organisation is a resource conversion machine: it takes inputs (labour, money, materials, equipment) from the outside world and converts them into useful products, goods and services — its outputs to customers.



External environment

All outside institutions and forces with an actual or potential interest or impact on the organisation's objectives: competitive, economic, technological, political, legal, demographic, cultural and ecosystem.

GLOBALISATION — AN OPPORTUNITY

Improving transport and communications let companies expand into worldwide markets — managers must handle multiple cultures and political systems.

Technology serving every level of the pyramid

The description distinguishes these types of information systems (any three):



TPS

Collects, stores, modifies and retrieves the organisation's transactions.



KWS

Supports technical expertise — word processing, CAD, expert systems.



MIS

Turns internal and external data into decision-making information.



DSS

Supports “what if” analysis and unstructured decisions.



ESS

A long-term, strategic view for senior management.

Transaction Processing Systems (TPS)

- A TPS collects, stores, modifies and retrieves the transactions of an organisation.
- A transaction is an event that generates or modifies data that is eventually stored in an information system.
- To be considered a transaction processing system, the computer must pass the ACID test.
- Batch processing is NOT transaction processing — several transactions are processed together and results are not immediately available while a transaction is entered.

Features of a TPS



Rapid response

Customers cannot be kept waiting — input to output in a few seconds or less.



Reliability

A breakdown can stop the business — very low failure rate, quick recovery, solid backups.



Inflexibility

Every transaction is processed the same way — no non-standard operations per user or agent.



Controlled processing

Enforces the organisation's allocated roles and responsibilities.

Four properties every transaction must have



Atomicity

All or nothing: an R5,000 transfer must debit the sender AND credit the receiver — or be cancelled entirely.



Consistency

Follows the rules and keeps the data correct: sender R20,000 → R15,000, receiver R10,000 → R15,000.



Isolation

Simultaneous transactions do not interfere: Client A's R5,000 and Client B's R2,000 each run as if on their own.



Durability

Once the transfer completes, the balances are saved permanently — a crash or power loss cannot undo them.

Databases and files

A database is a collection of data neatly organised, storing the accounting and operational records — accurate, protected, and used many times a day.



Hierarchical

Data in a series of levels — top-to-bottom nodes and branches; each child links to only ONE parent node.



Network

Nodes and branches like hierarchical — but each child can link to MULTIPLE higher parent nodes.



Relational

Data in a series of related tables — flexibility comes from the relationships built between tables.

Batch and on-line systems



Batch systems

Transactions are collected over a period and dealt with in one go — mail-order tickets batched in 50s, keyed in, then the stock-control program runs. Hours may pass, but a 4–5 day delivery makes that acceptable.



On-line systems

Updates happen immediately — critical for production-line sensors checking imbalances. A travel-agent ticket updates the airline's computer within seconds: pseudo-on-line processing.

Software for technical expertise

- Word-processing for clerical staff · spreadsheets for accounts and sales · database management systems for records.
- CAD for designers · project management systems · expert systems for specialist staff.
- Expert system example: an engineer types in the parameters for a bearing and the system suggests several alloys — the engineer's knowledge and experience make the final choice.
- Computers are networked; large organisations link several LANs so staff communicate by e-mail, document scanning, web-cams and video conferencing.

The MIS and the manager's five functions

The role of an MIS is to convert data from internal and external sources into information used to make effective decisions for planning, directing and controlling.



Planning — the direction a company takes, e.g. diversifying, where to operate



Organising — resources such as people, space, equipment and services



Coordinating — the activities of various departments



Decision-making — the organisation, products, employees, use of IT



Controlling — monitoring and supervising the activities of others

From data processing to management information



Data processing systems

Record day-to-day transactions — e.g. the sale of a CD to a customer.



Operational information systems

Read the collected data and produce lists — e.g. items that need to be re-ordered.



The MIS

Analyses sales data to highlight trends per product line — special promotion, or discontinue?

Information sources and flows

- Internal information comes easily from systems on the company network, e.g. sales figures per product line.
- External information: intelligence about competitors (press, leaks, even industrial espionage) · population shifts · economic and social factors · government legislation.
- Formal flows follow a procedure — Monday-morning branch downloads, agencies like Dun and Bradstreet, shared documents, e-mail, company intranets.
- Informal flows come from chance meetings, magazines, newspapers, or the news on TV.

Three levels of information for managers

Level	Decisions it serves	Example
Operational	Day-to-day decisions	Ordering in more stock
Tactical	Short to medium term	Introducing a new product to a retail outlet
Strategic	Long term — the organisation's future	Opening a new store, or taking over a rival

TYPES OF DECISION

Structured, unstructured — and the four stages



Structured

Repetitive, with a definite routine — stock below 15% triggers an order.



Unstructured

Needs knowledge, insight and evaluation — may crop up without warning.

THE DECISION STAGES

1. Recognise the problem — the MIS shows where performance is off.
2. Consider the solution — “what if” scenarios in a spreadsheet.
3. Choose — the manager's experience plus the MIS's information.
4. Implement and review — the MIS supplies the evaluation data.

Your work for Lesson 2



Questioning session

System concept, Input–Process–Output, three types of IS — AI-marked.



Knowledge check quiz 2

10 questions on systems theory and information systems. 80%+ is competent.



Self assessment

Tick the systems-theory statement honestly — or write it down as a goal.



Logbook project — Research

Your IT-in-business research project also evidences this outcome. Mark it 114050.



Every business is a system — IT keeps it running.

Inputs → process → outputs · TPS, KWS and MIS · ACID-safe transactions · databases · batch and on-line processing · information for operational, tactical and strategic decisions.